

BANKING & LOAN RISK ANALYTICS

Customer & Loan Default Risk Analysis

Project Focus	Loan default patterns, customer risk factors, SQL analytics and Power BI reporting
Dataset	Home Credit Default Risk – application_train and related credit/application tables
Primary Target	TARGET (0 = repayment / non-default, 1 = payment difficulties / default)

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1. Executive Overview

This project analyses loan application data to understand observed default risk across customer demographics, financial characteristics, employment, housing, education and loan-related variables. The workflow progresses from data understanding and quality checks through PostgreSQL database design, SQL cleaning and business analysis, and finally Power BI dashboard reporting.

The main analytical objective is not to build a predictive model in this report, but to identify groups with different observed default rates and provide evidence-based areas for further credit-risk assessment.

Key portfolio indicators

KPI	Value
Total applications	307,511
Total defaults	24,825
Overall default rate	8.07%
Non-default applications	282,686
Non-default share	91.93%

The Power BI dashboard supplied with the project reports the same headline portfolio figures: 307,511 total applications, 24,825 defaults and an 8.1% displayed default rate.

2. Project Objectives

- Understand the structure and quality of the Home Credit Default Risk dataset.
- Validate row counts, primary keys, duplicates, missing values and core business fields.
- Design a relational PostgreSQL structure using staging and warehouse concepts.
- Clean and validate the application data using SQL.
- Analyse observed default rates across customer and loan characteristics.
- Identify higher-risk segments for further credit-risk assessment.
- Present the findings through an interactive Power BI dashboard.

3. Dataset & Data Understanding

The project uses the Home Credit Default Risk dataset. The main application table is `application_train`. The primary customer identifier is `SK_ID_CURR` and the target variable is `TARGET`.

Dataset characteristics

Item	Finding
Application records	307,511
Primary key	SK_ID_CURR
Target	TARGET
Target = 0	Customer successfully repaid / non-default
Target = 1	Customer experienced payment difficulties / default
Duplicates	0 duplicate customer IDs / duplicate records identified
Data types	Integer, float and object/categorical fields
Missing values	Present in multiple columns, especially housing/property variables

The data-understanding phase found no duplicate records, while missing values required treatment before downstream analytics. The target variable is imbalanced, with substantially more non-default than default cases.

4. Tools & Technologies

Technology	Use in Project
Python / Pandas	Data loading, inspection, descriptive analysis and exploratory analysis
Matplotlib / Seaborn	Exploratory visualisation
Jupyter Notebook	Structured analytical workflow and documentation
PostgreSQL	Relational database design, SQL cleaning and business analysis
SQL	Aggregation, filtering, validation, grouping, ratios and risk-factor analysis
Power BI	Interactive KPI and default-rate dashboard

5. Project Workflow

Raw CSV Files → Staging Schema → SQL Data Cleaning → Warehouse Schema → Exploratory Analysis → Business SQL Analysis → Power BI Dashboard → Business Recommendations

6. Database Design

The database-design phase defines three schemas: banking for business-ready analytical tables, staging for raw imported CSV data, and warehouse for cleaned and transformed reporting tables.

The project documents seven principal related datasets: application_train, bureau, bureau_balance, previous_application, installments_payments, POS_CASH_balance and credit_card_balance. Key identifiers include SK_ID_CURR, SK_ID_PREV and SK_ID_BUREAU, with documented relationships between the customer/application and bureau datasets.

The staging approach initially imports columns as TEXT to reduce import failures caused by inconsistent source formats. Data types are then assigned during cleaning and warehouse preparation.

7. SQL Data Cleaning & Validation

The SQL cleaning phase validates the imported application data before analytical use.

Validation	Result
Row count	307,511 expected and 307,511 imported — passed
Duplicate customer IDs	0 — passed
TARGET values	Validated as 0 / 1
Gender values	Validated
Contract types	Validated
Negative income	Validated
Credit amount	Validated
Goods price	Validated
Children count	Validated
Family members	Validated
Missing values	Reviewed for key business columns

The documented validation process supports reliable downstream analysis. The project specifically notes that missing values are concentrated in several housing/property-related variables and that small sample categories should be interpreted cautiously.

8. Key Performance Indicators

Indicator	Value
Total applications	307,511
Total defaults	24,825
Default rate	8.07% (8.1% displayed on dashboard)
Total non-defaults	282,686
Non-default rate	91.93%

9. Key Findings

Gender: Male applicants have a higher observed default rate (10.14%) than female applicants (7.00%).

Age: Applicants under 25 have the highest observed default rate at 12.29%; the rate decreases across older age groups, reaching 5.22% for 55+.

Income: The 100K–200K income group has the highest observed rate among the analysed income groups at 8.55%; the 300K+ group has the lowest at 5.95%.

Credit amount: The 300K–500K credit group has the highest observed default rate at 9.93%.

Annuity: The 20K–40K annuity group has the highest observed default rate at 8.94%.

Goods price: The 300K–500K goods-price group has the highest observed default rate at 10.40%.

Credit-to-income: The 3–4 ratio group has the highest observed default rate at 8.85%, followed closely by 2–3 at 8.70%.

Credit-to-goods price: The 150%+ group has the highest observed rate at 12.93%; the 125%–150% group is also high at 12.10%.

Employment: Applicants with below 2 years of employment have an observed default rate of 8.66%, compared with 5.40% for 20+ years.

Loan term: The estimated Below 2 Years group has an observed default rate of 8.81%, versus 6.56% for 2–4 Years.

Education: Lower-secondary applicants have the highest observed rate among the main education categories at 10.93%; higher education is 5.36%.

Housing: Rented-apartment applicants have a 12.31% observed default rate, while applicants living with parents have 11.70%.

Occupation: Low-skill laborers have the highest observed occupation rate at 17.15%; drivers and waiters/barmen staff are also above 11%.

Family status: Civil marriage (9.94%) and single/not married (9.81%) show relatively higher observed rates among major family-status categories.

Important interpretation note: these are observed group-level default rates, not causal effects or a predictive credit score. Several very small groups are explicitly flagged in the notebooks as unsuitable for strong conclusions.

10. Power BI Dashboard

The supplied dashboard presents the loan portfolio through headline KPIs and default-rate breakdowns. The dashboard includes default versus non-default, gender, age group, income group, credit amount, annuity amount, employment duration and loan-term analyses, with slicers for demographic and housing/education attributes.

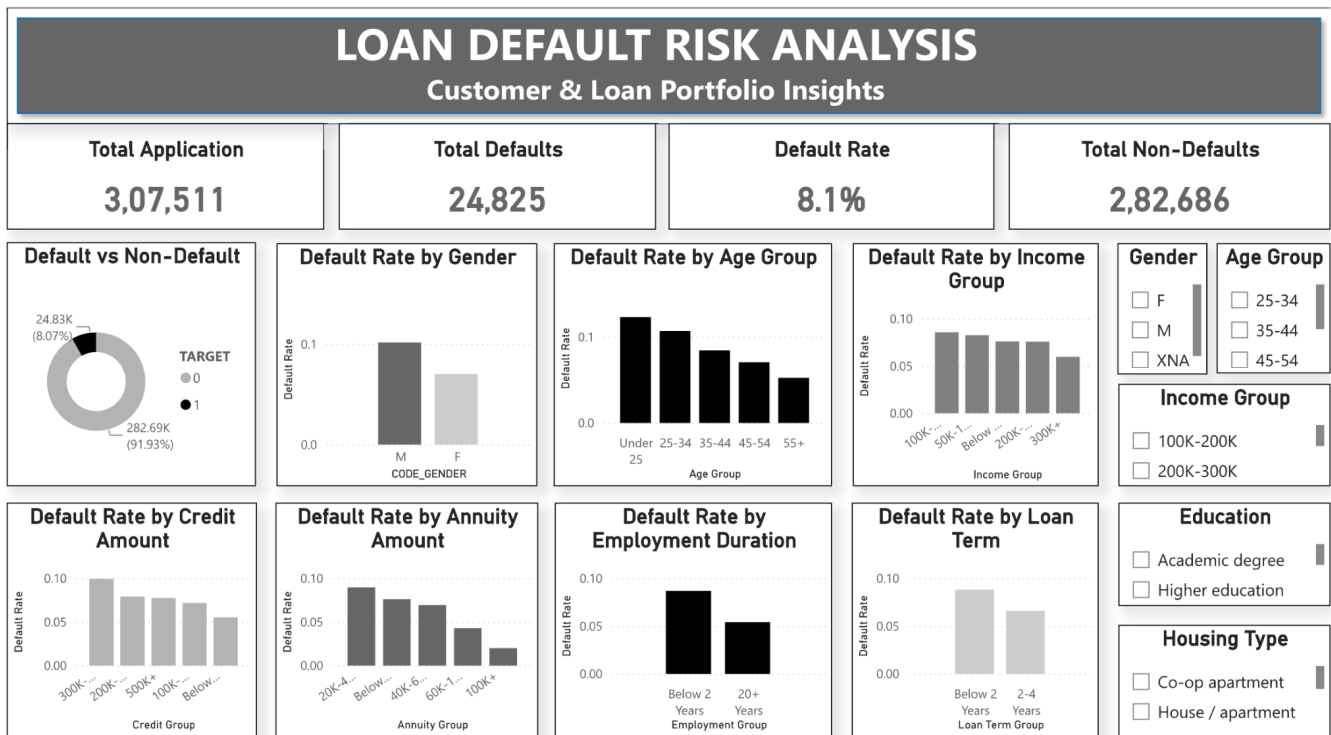


Figure 1. Loan Default Risk Analysis dashboard supplied with the project.

11. SQL Business Analysis

The SQL business-analysis phase converts the cleaned application data into decision-oriented comparisons. The analysis uses CASE expressions to create business groups, aggregation to calculate application/default counts, and ratios to calculate default rates.

Main analytical areas

- Default rate by gender and age group
- Default rate by income, credit amount and annuity amount
- Default rate by goods price
- Credit-to-income and annuity-to-credit ratios
- Credit-to-goods-price ratio
- Employment duration and estimated loan term
- Education, housing, occupation and family-status comparisons
- Overall portfolio baseline and final risk-factor summary

12. Business Recommendations

Based directly on the observed patterns documented in the project:

- Use age and employment duration as supporting risk-segmentation variables, particularly when assessing younger or shorter-tenure applicants.
- Review applications with high credit relative to goods price, especially ratios above 125%, because the observed default rates are materially higher in those groups.
- Give additional attention to the 300K–500K credit range and 300K–500K goods-price range, which show comparatively high observed default rates.
- Use credit-to-income ratio as a supplementary affordability indicator; the 2–4 range shows relatively elevated observed default rates.
- Treat income, education, housing and occupation as complementary risk dimensions rather than relying on a single characteristic.
- Avoid strong conclusions from tiny categories; validate segment volume before using a rate for business decisions.
- Use the Power BI dashboard to monitor default rates and application volumes across the major customer and loan dimensions.

13. Skills Demonstrated

- Data understanding and profiling
- Data quality validation
- Missing-value assessment
- Python / Pandas exploratory analysis
- Data visualisation
- PostgreSQL database design
- Relational schema and key design
- SQL data cleaning and validation
- SQL aggregation and segmentation
- Risk-factor and ratio analysis
- Power BI dashboard development
- KPI design and business storytelling
- Data-driven business recommendations

14. Project Learning Outcomes

This project demonstrates an end-to-end analytics workflow rather than a single isolated technique. It shows how raw financial data can be inspected, validated, structured in a relational database, analysed with SQL, explored with Python and communicated through Power BI.

- Understand how data-quality issues can affect financial analytics.
- Translate business questions into measurable default-rate metrics.
- Build a staging-to-warehouse database workflow.
- Use SQL to turn raw records into business segments and risk indicators.
- Interpret imbalanced target distributions appropriately.
- Communicate analytical findings through dashboard KPIs and visual comparisons.
- Recognise the difference between observed association and predictive/causal conclusions.

15. Repository Structure

The planned GitHub repository is organised into DATA, NOTEBOOK, SQL, Power BI, IMAGES and REPORTS folders. Because the source CSV files exceed GitHub's standard browser upload limit, the repository can document the datasets in DATA/README.md while keeping the large raw files outside the repository.

16. Conclusion

The Banking & Loan Risk Analytics project establishes an analytical baseline for understanding observed loan-default risk. Across 307,511 applications, 24,825 are recorded as defaults, giving an overall observed default rate of 8.07%. The analysis identifies meaningful differences across age, gender, income, credit amount, annuity amount, employment duration, loan term, education, housing, occupation and financial ratios.

The strongest observed signals include higher default rates among younger applicants, shorter employment duration, certain credit and goods-price ranges, and higher credit-to-goods-price ratios. These findings are suitable for portfolio segmentation and further modelling, while small-sample categories should be treated cautiously.

17. About the Author

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Project: Banking & Loan Risk Analytics

Report prepared from the project's supplied notebooks, SQL workflow documentation and Power BI dashboard.